Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2024	ENGI3491-WE01

Title: (BEng/MEng)

Digital Electronics & Digital Signal Processing

Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Question 1

a) An electronics manufacturer requires an asynchronous sequential logic network 'counter' circuit which meets all of the following specifications:

- The network has two external digital inputs, U and D, which are initially both set to 0.
- The network has two external digital outputs, Y and Z, which are initially both set to 0.
- From this initial 'reset' state:

  If U changes first then:

  On each rising or falling edge of U thereafter, the network will count up in Gray code order and roll over to YZ = 00 after YZ = 10. This progression is independent of any further changes in D, until the network returns to its 'reset' state where YZ = 00.

  If D changes first then:

  On each rising or falling edge of D thereafter, the network will count down in Gray code order, i.e. after the first rising edge, YZ rolls back to 10 and then to YZ = 11 after the next edge. This progression is independent of any further changes in U until the network returns to its 'reset' state where YZ = 00.
- Once the network has returned to the 'reset' state then either up or down counting can start again, in line with the above-described behaviour, depending on which external input changes first.
- The behaviour for when both U and D change to 1 whilst the system is in the 'reset' state, at the same time, is undefined.

continued

i) Draw a state diagram which describes the behaviour of a network which meets all of these specifications. State any reasonable assumptions.

ii) Design an asynchronous sequential logic network based on the state diagram that you have produced in part i).

Your answer should include all relevant working including a primitive flow table and a complete set of Boolean equations that could be used to implement the network directly. Clearly state any reasonable assumptions. Note that the Boolean equations do not need to be fully simplified and you do not need to produce a circuit diagram.

iii) With reference to the fundamental mode model, briefly explain why it might be reasonable to not define the behaviour in the specification when both U and D change to 1 from the reset condition. State a disadvantage that might arise from not explicitly defining this behaviour.

iv) The manufacturer has developed an ultra-high precision switching system for the U and D inputs which means that these inputs could, in principle, change at exactly the same time. Describe how your network would behave if this happened by considering the states that your network might enter. **[70%]**

b) Describe the advantages and disadvantages of the asynchronous sequential logic approach that you used in part a) versus a synchronous implementation. What additional digital components and signals would be required for an equivalent synchronous design? **[30%]**

continued

Question 2

A manufacturer has a requirement for an iterative logic network circuit which is capable of detecting a 6 bit sequence of either all 1s or all 0s. An output, Z, is high when either sequence has been detected and otherwise it is low.

a) i) Design an iterative logic network which meets the above specification. Show all relevant working and state any reasonable assumptions. You should produce all the simplified Boolean expressions which are necessary to implement the network using two input AND / OR gate logic only (with inverters if necessary).

ii) Show how equivalent behaviour to the iterative network that you designed in part i) can be achieved with a combinational logic network using two input AND / OR gate logic only (with inverters if necessary). Produce a Boolean expression which could be used to implement the network.

iii) Assume that the AND gate and OR gate propagation delays are both 2 ns and inverter propagation delays are 1 ns. Calculate the worst-case propagation delay for the networks that you have designed in parts i) and ii).

[50%]

b) Write VHDL code which describes the iterative logic network that you designed in part a) i). State any assumptions and use std_logic and a structural coding style.

[50%]

END